Partner-Specific Affective Precision in Social Active Inference

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Abstract. In multi-agent social settings, behavior and model reliability vary significantly across relationships. Beyond accurately inferring what other agents will do, an agent must also calibrate how confidently to act on those inferences—and do so independently for each relationship. An agent may carry a well-validated model of one partner, a fragile model of another, and an actively revising model of a third; collapsing these into a single confidence estimate loses information that matters for action. Affect has traditionally been described in terms of valence and arousal, but it can also serve a functional role: regulating how strongly beliefs are expressed in action rather than shaping their content. Using the active inference framework, we model affect as a metacognitive estimate of confidence in one's current partner model, and formalize this by extending affective precision to be relationship-specific. Each partner's behavioral evidence updates its own confidence estimate, which modulates how strongly the agent commits to its inferred partner state during action selection.
Simulations in a multi-partner trust game show that partner-local affective precision tracks predictive reliability rather than realized value: a reliably cooperative and a reliably defecting partner can both support high precision, because what they share is predictability, not positive value. In open decision regimes, behavioral effects arise through action sharpening rather than improved partner-state inference. Under abrupt behavioral change, accumulated confidence reshapes action deployment in ways that depend on portfolio structure and reallocation speed, indicating that affective precision is a calibration mechanism rather than a guaranteed payoff-improvement mechanism. Varying precision gain and prior model fitness reveals a family of trust-calibration profiles, providing a computational basis for individual differences in social trust.
Together, these results describe affective precision as a relationship-specific metacognitive signal that separates social prediction from social commitment.

Keywords: Active inference · Affective precision · Social metacognition · Multi-agent systems · Trust · Individual differences

1 Introduction

Social decision-making involves three problems that are useful to separate computationally. The first is inferring what a partner is like—whether they tend to cooperate, defect, or behave unpredictably. The second is modeling what they know or intend—the theory-of-mind question of how another agent represents the world and plans action [9, 10, 16]. The third is estimating whether one's current partner model is reliable enough to guide confident action. Accurate type inference and sophisticated mentalizing do not by themselves solve this third problem. Knowing that someone is unreliable is not the same as knowing how cautiously to treat them; knowing that someone is predictable is not the same as knowing how strongly to commit to that prediction. An agent that infers accurately but commits poorly wastes its social knowledge; one that commits confidently on fragile models exposes itself to exploitation. Many formal models specify what agents infer about others while treating confidence in those inferences as a fixed or global quantity.

We model social affect as a metacognitive estimate of whether the focal agent's current partner model warrants confident action, rather than as a direct readout of another agent's future behavior or a record of reward and punishment. In psychology and neuroscience, affect is often described in terms of valence and arousal—the felt pull toward or away from situations and social commitments [2]. Functionally, however, it can regulate how strongly beliefs are expressed in action: affect assigns action-relevant weight to what an agent already infers, shaping the degree of commitment rather than the content of inference.

The active inference framework [7, 8, 15] provides a formal route from this observation to a computational mechanism. In active inference, action selection is precision-weighted: the same beliefs can produce decisive or tentative behavior depending on γ, the parameter controlling how sharply an agent commits to its preferred policies. Hesp et al. [12] made affect endogenous by modeling it as inference over a higher-order belief β representing subjective model fitness—an affective charge signal ϕ drives updates to β, rising when the model performs better than a prior baseline and falling when it performs worse. On this view, affect is metacognitive rather than mentalistic: it tracks model fitness rather than partner intentions or reward history alone. Recent work on theory of mind in active inference [16, 17] addresses how agents model what others know or intend through recursive belief structures and factorized generative models—solving problem two well, but assuming rather than estimating action confidence. Hierarchical volatility models [14, 3] modulate learning rate under uncertainty, a related spirit but operating on belief updating rather than action commitment.

What none of these approaches address is confidence at the level of individual relationships. In multi-partner settings, model fitness can differ sharply across partners—a person can carry a well-validated model of a longtime collaborator, a fragile model of a new acquaintance, and an actively revising model of someone who has recently behaved unexpectedly. Collapsing these into a single shared precision budget loses information that matters for action. We therefore extend affective precision to be relationship-specific: each partner receives its own β estimate, updated only from that partner's behavioral evidence, which modulates policy precision when acting toward that partner.

We evaluate this mechanism in a repeated multi-partner trust game across binary and graded decision regimes, partner-choice conditions, abrupt betrayal, and profile-style parameter variation. The simulations evaluate three predictions: that partner-local affective precision tracks predictive reliability more than realized value; that behavioral effects arise through action sharpening rather than improved partner-state inference; and that precision gain together with prior model fitness shapes temporally structured profiles of social trust revision.

The remainder of the paper is organized as follows. Section 2 specifies the multi-partner trust game, per-partner generative models, and the partner-local affective precision mechanism. Section 3 reports simulation results. Section 4 discusses implications, limitations, and directions for future work.

2 Methods

A focal active-inference agent plays a repeated trust game with four partners whose behavioral types and stances are hidden and must be inferred from behavioral evidence. The agent is implemented using the `pymdp` library for discrete active inference [11, 6], which provides standard components for belief updating and policy selection under the active inference framework [7, 8, 15]. Partner-local affective precision is layered on top of per-partner generative models, allowing evidence and action confidence to remain relationship-specific. Full POMDP specifications, affective precision update rules, and simulation protocols are provided in Appendix A, Appendix B, and Appendix C, respectively.

2.1 Trust game

The trust game [4, 13] provides a natural testbed because partner types are hidden and must be inferred, creating a genuine gap between social knowledge and actionable confidence. Each partner has a latent *type*—cooperator, reciprocator, exploiter, or random—and a latent *stance* toward the agent—trusting, neutral, or hostile—neither directly observable; both must be inferred from behavioral evidence. Episodes run for up to 200 rounds.

In the binary game both players choose to cooperate or defect, with focal agent payoffs as shown in Table 1. The structure poses a social inference problem: defection is locally tempting, but cooperative partners are worth identifying and maintaining over repeated interaction. In the graded regime, the binary choice is replaced by a discretized investment level. In the partner-choice regime, the agent selects which partner to engage before choosing an action each round, making approach and avoidance explicit.

Table 1: Focal agent payoffs in the binary trust game.

	Partner cooperates	Partner defects
Agent cooperates	3	−1
Agent defects	5	1

Reported experiments use one focal active-inference agent interacting with four environment-side partners. Partners sample actions from type-by-stance cooperation probabilities and undergo action-dependent stance transitions, but do not perform variational inference or maintain affective precision estimates. This isolates the focal agent's precision mechanism before extending to reciprocal active-inference partners.

2.2 Per-partner generative models

Rather than maintaining a single pooled social model, the focal agent maintains a separate discrete generative model $\mathcal{M}_k$ for each partner k. This factorization keeps evidence about each partner independent and allows action confidence to vary across relationships. Each model encodes observation likelihoods, transition dynamics, preferences, priors, and policy priors using the standard $A/B/C/D/E$ structure of discrete active inference [6]. Full matrix specifications are provided in Appendix A.

Each partner model contains three latent factors: partner type, partner stance, and a deterministic own-action bookkeeping factor that makes the agent's executed action available to the payoff likelihood. Observations are partner action and focal payoff. The partner-action likelihood is defined by type- and stance-conditioned cooperation probabilities, operationalizing predictive reliability: both a reliably cooperative and a reliably defecting partner can be predictable, even though their payoff consequences differ. Payoff enters as a second observation modality, with partner action marginalized through the cooperation likelihood, so preferences over outcomes remain part of the generative model rather than an external reward cache. Stance transitions are action-dependent—cooperation tends to build trust, defection damages it, and recovery from hostility is slower than collapse.

2.3 Partner-local affective precision

Hesp et al. [12] treat policy precision as an inferred quantity, modeling affect as active inference over a higher-order belief β representing subjective model fitness. We extend this to the per-partner case by assigning each partner its own β_k estimate, updated from that partner's behavioral evidence alone.

After interacting with partner k, the agent computes surprisal from the predicted probability of the observed partner response:

$$\epsilon_k = -\log P\big(o_k^{\text{act}} = \hat{o}_k^{\text{act}} \;\big|\; q(s_k^{\text{type}}, s_k^{\text{stance}})\big)\,. \tag{1}$$

This is computed from partner action rather than payoff because action is the most direct test of type-by-stance model fit: a reliably defecting partner produces no action surprise even though the payoff is poor. The neutral fitness baseline is set at $\sigma_0^2 = (-\log 0.5)^2$, so a fifty-fifty binary prediction carries zero affective charge. Affective charge is then:

$$\phi_k = \alpha\big(\sigma_0^2 - \epsilon_k^2\big)\,, \tag{2}$$

where α is a gain parameter. Positive ϕ_k means partner k's model predicted better than baseline; negative ϕ_k means it predicted worse. A categorical posterior $q(\beta_k)$ is updated each round by combining a slowly evolving persistence prior with a charge-based pseudo-likelihood; full update rules are given in Appendix B. The agent then sets policy precision from the posterior mean:

$$\gamma_k = \gamma_{\text{base}} \,/\, \mathbb{E}_{q(\beta_k)}[\beta_k], \tag{3}$$

where γ_{base} is the baseline policy precision applied in the absence of affective modulation. Here β_k represents inverse precision: higher β_k means the model has been less reliable, producing a broader, less committed policy distribution. Accordingly, positive model fitness is expressed as lower expected β_k and therefore higher policy precision γ_k; when we refer below to precision or confidence tracking reliability, we mean this β_k-derived policy-precision signal rather than raw β_k increasing with reliability. The mechanism thus translates accumulated evidence about predictive reliability into action confidence independently for each social relationship. The β_k posterior is maintained as an auxiliary precision tracker rather than a proper hidden state in the generative model, which simplifies implementation at the cost of losing the agent's ability to form prior expectations over its own future affective states; the implications of this design choice are discussed in Section 4.

2.4 Cross-partner policy selection

In partner-choice regimes, policies specify both which partner to engage and how to act toward that partner. A naive application of partner-local precision creates an artifact: low precision toward a uniformly poor partner branch scales its logits toward zero, making it spuriously attractive in cross-partner comparison.

We address this by applying precision to centered branch logits:

$$\tilde{u}_{k,\pi} = \bar{u}_k + \gamma_k \left(u_{k,\pi} - \bar{u}_k\right), \tag{4}$$

where $u_{k,\pi}$ is the pre-precision policy logit for policy π toward partner k and $\bar{u}_k$ is that branch's mean logit. Centering preserves the mean score of each partner branch while allowing γ_k to sharpen or flatten the within-partner policy distribution. The agent selects from the combined set of centered logits across all partner branches. Full details are provided in Appendix B.

2.5 Simulation setup

Simulations compare four affect-runtime variants: full partner-local precision, a shared-β ablation that pools evidence across partners while keeping per-partner POMDP beliefs intact, a tracked-only ablation that updates $q(\beta_k)$ but fixes $\gamma_k = \gamma_{\text{base}}$, and a no-affect control that disables the β tracker entirely. Conditions span binary and graded decision regimes, partner-choice settings, abrupt betrayal, and profile-style parameter variation across precision gain α and prior model fitness. Central comparisons use 30 seeds per condition; profile-scale experiments use 20 (Appendix C). Full condition definitions, episode lengths, and metrics are provided in Appendix C. Simulation code, analysis scripts, and experiment configurations will be released in a public repository upon acceptance.

3 Results

The experiments are organized around three dissociations. The first tests whether β_k-derived policy precision tracks predictive reliability independently of realized value, and whether pooling evidence across partners destroys this relationship-specific signal. The second tests whether precision affects behavior through action commitment rather than improved partner-state inference. The third examines how the mechanism behaves under social change, and how precision gain and prior model fitness shape the temporal structure of trust revision. Each dissociation is evaluated by varying one element of the design while holding others fixed. We therefore treat payoff as a downstream diagnostic of how confidence calibration interacts with task structure, not as the primary criterion for whether the mechanism is present.

3.1 Precision tracks predictability, not partner value

The affective charge formula uses action surprisal rather than payoff, so the inverse-precision posterior $q(\beta_k)$ moves with behavioral consistency rather than realized value: reliable predictions shift mass toward lower β_k and higher γ_k, while unreliable predictions shift mass toward higher β_k and lower γ_k. Once the agent has inferred a partner's type and stance, a reliably defecting partner can generate low surprisal despite poor payoff, while an erratic partner can generate high surprisal despite occasionally favorable outcomes. The precision signal is therefore expected to track the quality of the agent's predictive model, not the desirability of the partner's behavior.

The 30-seed model-fitness analysis separated reliability from value. Across partner-seed units, the β_k-derived precision signal shows a stronger absolute association with action surprisal than with realized payoff (0.949 versus 0.748). Because payoff, surprise, and exposure are correlated in partner-choice episodes, we also computed an active-encounter partial readout: after controlling encounter imbalance, the partner-local precision–surprise association remains strong (0.882), whereas the precision–payoff association falls to 0.130. A single shared β tracker

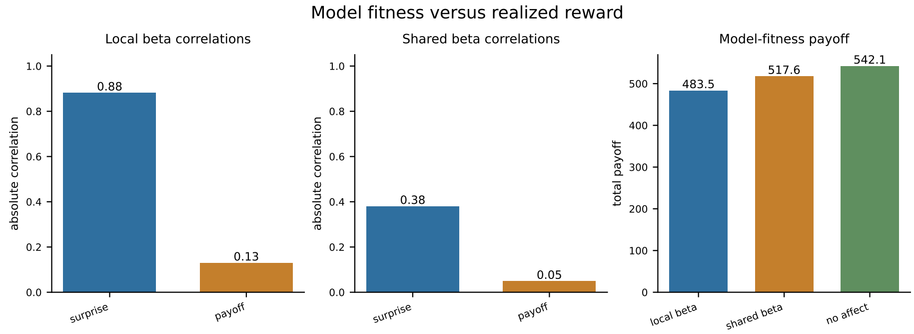

Fig. 1: Partner-local precision tracks action surprisal more strongly than realized payoff. The panel contrasts active-encounter partial associations between mean partner precision and surprise versus payoff for partner-local β_k and a shared β tracker, plus aggregate payoff. Partner-local tracking preserves the surprise-over-reward dissociation; pooling evidence across partners attenuates relational specificity.

preserves the ordering but with weaker relational specificity (partial associations 0.380 versus 0.049). Aggregate payoff followed a different pattern (partner-local: 483.5; shared: 517.6; no-affect: 542.1), separating the precision signal from reward maximization. Partner-local β_k-derived precision tracked predictive model fitness more closely than realized partner value. Figure 1 visualizes this dissociation for partner-local versus shared precision tracking.

3.2 Shared precision tests relational specificity

The reliability-over-value dissociation in Section 3.1 depends on evidence remaining tied to the partner who generated it. The shared-β ablation tests this directly: if pooling evidence across partners destroys the partner-specific signal, it confirms that locality is doing the work rather than the update rule alone. A shared β tracker preserves the coarse ordering between predictive surprise and realized payoff, but attenuates the partner-specific dissociation observed under partner-local precision. In the active-encounter partial readout, partner-local precision shows a stronger association with surprise than payoff, whereas shared precision compresses both associations toward weaker values.

Locality is therefore load-bearing: pooling evidence blurred model-fitness estimates across relationships, while partner-local $q(\beta_k)$ preserved the link between each partner's behavioral evidence and that partner's β-derived action confidence. Whether this preservation translates to behavioral advantage depends on the task structure, as the mixed-volatility analyses below show.

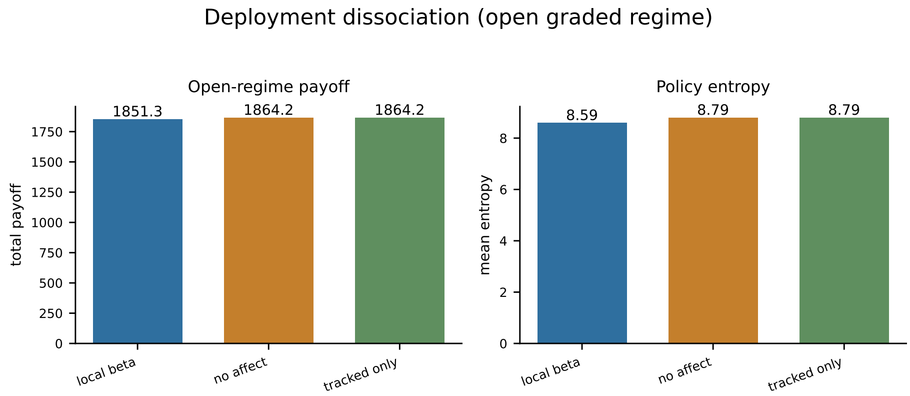

Fig. 2: Open graded regime: partner-local affective precision lowers policy entropy relative to no-affect and tracked-only controls without a corresponding payoff advantage. Tracked-only updates β_k but fixes γ_k; its deployment readouts match the no-affect baseline, isolating the entropy effect to the $\beta_k \to \gamma_k$ action-confidence pathway.

3.3 Behavioral effects arise through action sharpening

A tracked-only ablation updates $q(\beta_k)$ normally but fixes $\gamma_k = \gamma_{\text{base}}$, allowing the precision estimate to be analyzed without letting it affect action selection. In the open graded decision regime, full partner-local precision lowers policy entropy relative to both baselines (8.6 versus 8.8 for no-affect and tracked-only), while total payoff is flat-to-slightly lower (1851.3 versus 1864.2 for both baselines). Tracked-only remains near the no-affect baseline, while full precision modulation lowers entropy. The behavioral effect therefore enters through the $\beta_k \to \gamma_k$ deployment pathway rather than through improved partner-state inference. This is the central intended effect of the mechanism: affective precision changes how strongly existing beliefs are expressed in action, even when that sharper deployment does not immediately improve cumulative payoff. Figure 2 summarizes the three-way contrast across affect, no-affect, and tracked-only variants. The next question is whether this action-sharpening effect extends to partner selection, not just action selection within a fixed partner.

3.4 Partner choice shifts under precision modulation

In partner-choice regimes, precision modulation can affect both action selection and partner selection. Policy entropy under precision modulation is 4.66 versus 4.81 without it, and engagement shows a modest redistribution across partner types. The partner-local agent selected the cooperator on 36.6% of rounds and the reciprocator on 31.6%, while allocating 13.8% and 18.0% of choices to the exploiter and random partners, respectively. The no-affect control selected

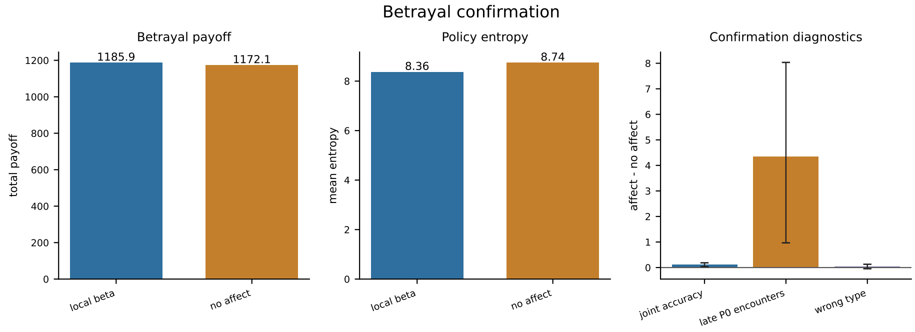

Fig. 3: Abrupt betrayal in the partner-choice regime (30 seeds). Partner-local affective precision lowers policy entropy relative to the no-affect control; the payoff difference is positive but small, and its bootstrap interval crosses zero. Joint partner-type accuracy is also higher under precision modulation, indicating deployment change with modest behavioral benefit.

the same partners at 34.8%, 31.8%, 16.2%, and 17.2%, respectively. Aggregate payoff remains similar at this scale (377.3 versus 385.3), showing that selection reorganization can occur without an immediate cumulative-payoff advantage.

These allocation readouts show modest partner-choice redistribution without a clear payoff separation at this replication scale. Whether stronger payoff consequences emerge depends on the stability of partners' behavior over time — a question the abrupt-betrayal condition addresses directly.

3.5 Abrupt betrayal tests confidence under change

When a previously cooperative partner changes stance mid-episode, the agent enters the post-switch period with confidence accumulated under a model that no longer matches the partner's behavior. In the 30-seed abrupt-betrayal analysis, partner-local affective precision produces numerically higher payoff than the no-affect control (1185.9 versus 1172.1), but the paired bootstrap interval for the difference crosses zero (-25.2 to 53.2). The clearest effect is lower policy entropy under precision modulation (8.36 versus 8.74; bootstrap interval for the difference -0.62 to -0.14), confirming that the action-confidence pathway is active.

Joint partner-type accuracy is higher under precision modulation (0.372 versus 0.266; bootstrap interval for the difference 0.034 to 0.185). Under abrupt social change, partner-local precision produced lower entropy and higher joint partner-type accuracy, while the payoff difference remained uncertain. The benefit of accumulated confidence depended on portfolio structure and the speed with which the agent could redirect commitment after change. This temporal dependence motivates the profile analyses below, which vary precision gain and prior model fitness to examine how the confidence-revision process differs across agents.

3.6 Precision dynamics as computational profiles of social trust calibration

We varied precision gain α and the prior over model fitness to examine how confidence-revision dynamics change across social contexts. These parameter settings provide computational analogues for empirically studied variation in social trust calibration [3, 1, 5], but they are not fitted to human behavior.

Initial parameter perturbations (Appendix D) show that flattened precision dynamics can reduce action commitment without improving payoff: the low-gain variant compresses β_k range to 0.24 and raises entropy to 8.9, near the no-affect and tracked-only baselines, whereas expanded-dynamics variants widen β_k movement without monotonic payoff gains. We therefore sweep gain and prior structure directly rather than treating trust calibration as a single high-versus-low axis.

Precision-gain axis. Sweeping α across $\{0.05, 0.1, 0.3, 0.5, 1.0, 2.0, 4.0, 8.0\}$ shows that β_k movement range increases monotonically with α in the betrayal regime. Low-α agents ($\alpha \leq 0.1$) show near-zero exploitation rates and Gini coefficients of 0.423 and 0.397, producing diffuse, slowly-updating engagement patterns. High-α agents ($\alpha \geq 4.0$) show Gini coefficients of 0.366 and 0.408 with recovery times of 28.8 and 13.0 rounds respectively—faster detection of behavioral change in some runs, but higher variance in recovery outcomes. Payoff does not rank α values monotonically across either regime (betrayal: range 1914.6–1986.2; open graded: 1906.3–1987.7), indicating that the behavioral consequence of precision-gain depends on task context rather than gain magnitude alone. Extended alpha-sweep diagnostics are in Appendix D.

Gain-prior profiles. Crossing prior belief (naive versus cautious) with α (low versus high) defines four behavioral profiles tested in the betrayal environment (20 seeds). Low-α profiles (naive-stubborn and avoidant-rigid) show substantially smaller β_k range (0.32–0.36) than high-α profiles (0.89–1.08), confirming that gain controls the magnitude of precision dynamics. Naive-stubborn shows the lowest Gini (0.304), reflecting maximally diffuse partner selection consistent with a slowly-updating confidence profile. Avoidant-rigid shows the lowest payoff (1889), as the cautious prior prevents confidence from building even toward reliably cooperative partners. The anxious-reactive profile shows the highest post-betrayal commitment errors toward the switched partner (post-betrayal P0 selection: 0.319; high investment rate: 0.121), reflecting rapid confidence accumulation that becomes a liability when a partner's behavior changes. The default agent achieves the highest payoff among these variants (1991), suggesting that payoff depends on calibration within the gain-prior space. Quadrant figures and supporting diagnostics are in Appendix D.

Forgiveness and trust repair. In the forgiveness paradigm, the betrayed partner reverts to cooperative/trusting behavior after round 120. The main pattern separates reengagement from restored confidence. No-affect and cautious-low-α

agents reengage most often after repair (selection rates 0.593 and 0.630), while default and naive-high-α agents reengage less (0.475 and 0.560). Payoff recovery is near baseline across all conditions (0.996–1.033), separating reengagement from payoff recovery rather than producing a simple "more affect means better forgiveness" pattern. Low gain produces muted β_k dynamics and rapid reapproach in this task (cautious-low latency 1.5 rounds; naive-low 4.6), whereas high-gain profiles show larger confidence swings but not a higher payoff-recovery ceiling. Trust repair separates willingness to sample the reverted partner from restoration of model-fitness confidence.

Mixed-volatility discrimination. The mixed-volatility environment places four partners with heterogeneous volatility in the same episode: a stationary cooperator (P0), a stationary exploiter (P1), an abrupt-switch partner (P2), and a gradual-drift partner (P3). Under partner choice, the agent must maintain stable engagement with P0 while updating away from P2 and P3 after their behavioral changes.

Table 2: Mixed-volatility metrics by precision profile (20 seeds).

Variant	Payoff	Discrimination	P0 false positive	β range
default	1976	0.121	0.422	0.933
high-α	2054	−0.089	0.484	1.049
low-α	1927	0.213	0.359	0.182
no-affect	1934	–	0.273	–

The false-positive rate—unwarranted confidence reduction toward the stationary P0—increases with α: high-α (0.484), default (0.422), low-α (0.359), no-affect (0.273). The discrimination index does not simply improve with gain: low-α gives the strongest discrimination score (0.213), while high-α is negative (−0.089) despite achieving the highest payoff. Figure 4 shows that payoff did not increase monotonically with gain. In this environment, high gain amplified reactivity and false positives toward stable partners, while payoff depended on how that reactivity interacted with partner allocation.

Together, the profile experiments show a family of confidence-revision profiles determined by gain and prior initial conditions. Across environments, confidence dynamics, partner allocation, and payoff dissociated rather than following a monotonic payoff ranking. Extended profile tables, alpha-sweep diagnostics, and forgiveness/trust-repair analyses are provided in Appendix D.

4 Discussion

The results show that separating social prediction from social commitment is computationally tractable and can produce measurable behavioral consequences

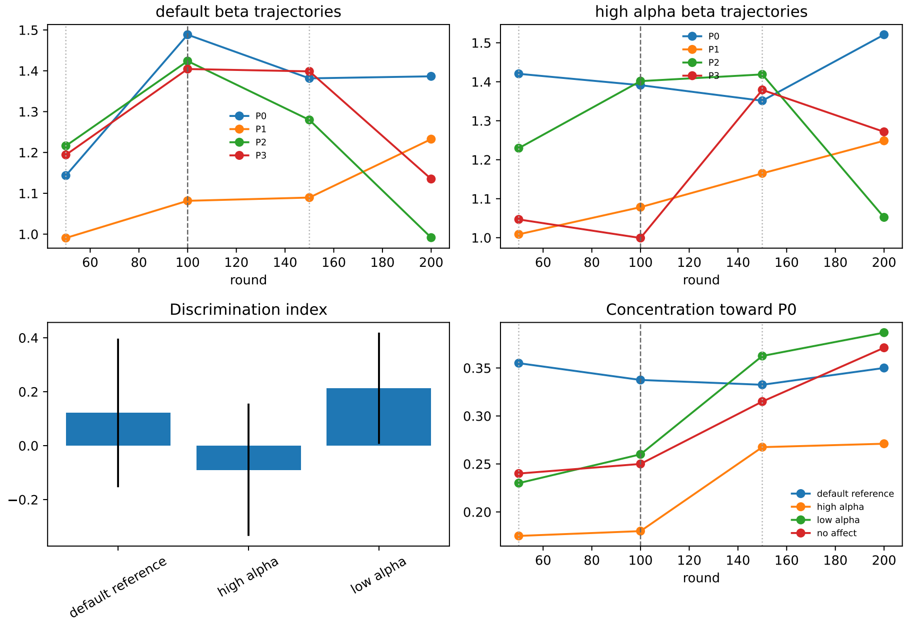

Fig. 4: Mixed-volatility partner-choice environment (20 seeds). Four partners differ in volatility: stationary cooperator (P0), stationary exploiter (P1), abrupt switch (P2), and gradual drift (P3). Higher precision-gain α can raise payoff while worsening discrimination and increasing false-positive confidence reduction toward P0.

under the present task designs. By giving each social relationship its own model-fitness estimate, the agent can know what a partner will do and separately regulate how strongly to act on that knowledge. The three main dissociations — reliability over value, deployment over inference, and partner-local over shared confidence — each isolate a different aspect of this separation. These dissociations should be interpreted as evidence for a confidence-calibration mechanism, not as evidence that the present implementation is already optimized for payoff maximization across all social environments. Payoff is treated here as a downstream consequence of calibration interacting with task structure, partner volatility, and reallocation speed.

The reliability-over-value dissociation follows from the update rule: $q(\beta_k)$ responds to action surprisal, not payoff, so predictability drives lower inverse precision and higher action confidence regardless of whether a partner is beneficial. This has a non-obvious implication — a reliably defecting partner and a reliably cooperative partner are computationally equivalent from the perspective of affective precision, because both are predictable. What the precision signal tracks is the quality of the agent's model, not the desirability of the partner's behavior.

This is the sense in which affect here is metacognitive rather than evaluative: it reflects how well the agent knows what is happening, not how good what is happening is.

The tracked-only ablation shows that the precision estimate only matters insofar as it reaches action selection. When β_k was updated but γ_k was held fixed, behavior was indistinguishable from the no-affect baseline. This rules out the possibility that the precision tracker improves partner-state inference as a side effect — the beliefs themselves are unchanged. What changes is the degree of commitment: the same inferred partner state produces sharper action under higher confidence. Affect here operates downstream of inference, on the mapping from beliefs to action rather than on the beliefs themselves.

The shared-β comparison shows why locality matters. Pooling evidence across partners preserves a coarse signal but loses the partner-specific structure — confidence becomes a property of the social environment rather than of individual relationships. In settings where partners differ in predictability, this is a loss of information that matters for action. Partner-local β_k keeps each relationship's evidence local to that relationship, which is an appropriate inductive structure when partners are genuinely independent. This is distinct from what theory-of-mind accounts in active inference address [16]: those approaches ask what another agent knows or intends, while partner-local affective precision asks how confidently to act on what one already infers. The two are complementary rather than competing.

The betrayal and mixed-volatility experiments reveal the temporal boundary conditions of the mechanism. Accumulated confidence is useful when the social environment is stable because it sharpens action toward partners whose behavior is well predicted. The same confidence can become a liability when a partner changes, because the model-fitness evidence was accumulated under a relationship that no longer has the same structure. The abrupt-betrayal results show this directly: the most consistent effect is deployment change rather than payoff improvement, and payoff consequences depend on how quickly engagement can be reallocated to alternative partners. The mixed-volatility results make the same point more sharply: higher gain can increase payoff in some settings while worsening discrimination and producing false-positive confidence reductions toward stable partners. This suggests that partner-local affective precision should be viewed as a first calibration layer, not a complete social-control architecture. Stronger performance gains will likely require additional mechanisms that decide when accumulated confidence should be discounted, when alternative partners should be sampled, and how quickly partner-choice policies should reallocate after detected change.

A natural next step is therefore to pair partner-local affective precision with volatility-sensitive confidence control. A change-detection or volatility layer could monitor shifts in prediction-error structure and discount stale confidence when a previously reliable partner begins producing systematically unexpected behavior. This would preserve the main benefit of the current mechanism — relationship-specific action confidence — while reducing the post-change liability of overcommitting to outdated partner models. In performance-oriented multi-agent systems, this would turn affective precision from a deployment modulator into part of a broader adaptive control loop: predict partner behavior, estimate confidence in that prediction, detect when confidence has become stale, and reallocate exploration or commitment accordingly.

The profile analyses show that social trust calibration does not vary along a single axis. Precision gain and prior model fitness jointly determine update speed, confidence amplitude, sensitivity to change, reengagement after repair, and vulnerability to false positives — and these dimensions trade off rather than co-vary. High gain can increase payoff while worsening discrimination; low gain produces stable engagement but slow recovery from betrayal; cautious priors suppress confidence even toward reliable partners. The profiles generate concrete behavioral predictions: an anxious-reactive agent (naive prior, high gain) should show rapid confidence accumulation followed by high post-betrayal commitment errors, while an avoidant-rigid agent (cautious prior, low gain) should show muted confidence dynamics and poor payoff even with cooperative partners. Whether these profiles correspond to stable individual differences in human social trust — connecting to work on social learning [3], psychosis [1], and attachment [5] — remains an empirical question that will require fitting the model to behavioral time series. For artificial agents, the same profiles also suggest engineering tradeoffs. High-gain agents may react quickly but risk false positives and post-betrayal overcommitment; low-gain agents may remain stable but fail to exploit reliable partners quickly; cautious priors may protect against premature trust while suppressing useful confidence. These tradeoffs are why the present paper does not treat any single profile as globally optimal.

Several architectural extensions would turn the present calibration mechanism into a more complete and performance-oriented social agent. First, β_k is currently maintained as an auxiliary tracker outside the generative model. Embedding it as a hidden state would allow the agent to form prior expectations over its own future confidence, anticipate changes in partner reliability, and choose policies partly by considering how confidence itself may evolve [12]. Second, a volatility or change-detection layer could discount accumulated confidence when prediction-error structure shifts, addressing the main limitation exposed by betrayal and mixed-volatility conditions. Third, partner-choice policies could be made more explicitly exploratory after detected change, allowing the agent to sample alternatives rather than simply sharpening or flattening action toward the currently selected partner. Fourth, environment-side partners in these simulations could be replaced with reciprocal active-inference agents that maintain their own beliefs, policies, and precision estimates. This would allow partner-local affective precision to interact with theory-of-mind inference in a fully reciprocal multi-agent setting.

Two empirical extensions follow from the profile results. The model produces behavioral quantities that can be measured from trust-game time series — exploitation sensitivity, betrayal recovery time, reengagement latency after repair, false-positive confidence reduction, and partner reallocation under mixed volatility. Fitting α and prior model-fitness parameters to individual participants would test whether the computational profiles correspond to stable individual differences in social trust calibration. The dissociations that are most robust at the current simulation scale — predictability over value, deployment over inference, and partner-local over shared confidence — are the natural targets for empirical validation first. Only after these dissociations are validated should future work ask whether richer versions of the architecture produce robust performance gains in more realistic social environments.

5 Conclusion

Predicting what a partner will do, modeling what they know or intend, and deciding how confidently to act on one's current partner model are separable problems. Partner-local affective precision addresses the third as a relationship-specific metacognitive confidence signal. The simulations show that it tracks predictive reliability more than realized value, affects action deployment through policy precision rather than improved partner-state inference, and creates temporally structured trust-revision profiles under social change, while payoff consequences depend on task structure and reallocation dynamics. By giving each social relationship its own model-fitness estimate, the mechanism allows action confidence to be calibrated independently across a social portfolio. Precision gain and priors over model fitness define a family of computational trust profiles for individual-difference research, providing a computational account of how social affect can regulate confidence in action. Future performance-oriented versions of the model should build on this calibration layer by adding explicit volatility detection, adaptive confidence discounting, and reciprocal partner models.

References

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Appendix

A Full POMDP Specification

This appendix specifies the POMDP components used by the focal agent.

A.1 Hidden state factors and observations

As described in Section 2.2, the focal agent's generative model for each partner k maintains three latent factors:

$$s_k^{\text{type}} \in \{\text{cooperator, reciprocator, exploiter, random}\} \quad \text{(latent behavioral type)} \tag{5}$$

$$s_k^{\text{stance}} \in \{\text{trusting, neutral, hostile}\} \quad \text{(latent stance)} \tag{6}$$

$$s^{\text{own}} \in \mathcal{A} \quad \text{(agent's executed social action)} \tag{7}$$

The own-action state is *deterministic bookkeeping*: the agent always knows its own action, and this factor makes that knowledge available to the observation model. It is not a psychological hidden variable. In the binary task, the joint hidden state per partner spans $4 \times 3 \times 2 = 24$ possibilities; graded regimes replace the binary action factor with the configured investment levels.

Observations are partner action and payoff:

$$o_k^{\text{act}} \in \{\text{cooperate, defect}\} \quad \text{(partner response)} \tag{8}$$

$$o_k^{\text{pay}} \in \{-1, 1, 3, 5\} \quad \text{(focal agent's realized payoff)} \tag{9}$$

A.2 Partner-action likelihood

The core of the generative model is the *partner cooperation likelihood table*, which defines the probability the partner cooperates given type and stance:

Table 3: Cooperation likelihood $P(\text{cooperate} \mid s^{\text{type}}, s^{\text{stance}})$.

Partner type	Trusting	Neutral	Hostile
Cooperator	0.95	0.80	0.55
Reciprocator	0.90	0.70	0.30
Exploiter	0.70	0.35	0.10
Random	0.60	0.50	0.35

This table is the foundation of the likelihood modality A_k^{act}. Entries define

$$P\left(o_k^{\text{act}} = \text{cooperate} \mid s_k^{\text{type}}, s_k^{\text{stance}}\right), \tag{10}$$

with $P(o_k^{\text{act}} = \text{defect} \mid \cdot) = 1 - P(o_k^{\text{act}} = \text{cooperate} \mid \cdot)$. The modality is uniform over s^{own}: partner action depends only on inferred type and stance. This operationalizes what the agent considers a predictable partner—a cooperative reciprocator in a trusting stance has high predicted cooperation; an exploiter in a hostile stance has low predicted cooperation.

A.3 Payoff likelihood

Payoff enters as a second observation modality $A_k^{\text{pay}} = P(o_k^{\text{pay}} \mid s^{\text{own}}, s_k^{\text{type}}, s_k^{\text{stance}})$ rather than as external reward. The environment payoff table is deterministic (Table 1 in the main text); in the generative model, partner action is marginalized through the cooperation likelihood:

$$P\big(o_k^{\text{pay}} = v \mid s^{\text{own}}, s_k^{\text{type}}, s_k^{\text{stance}}\big) = \sum_{a^{\text{partner}} \in \{\text{C},\text{D}\}} \mathbb{I}\big[\pi(s^{\text{own}}, a^{\text{partner}}) = v\big]\, P\big(o_k^{\text{act}} = a^{\text{partner}} \mid s_k^{\text{type}}, s_k^{\text{stance}}\big)\,, \tag{11}$$

where $\pi(\cdot,\cdot)$ maps joint actions to focal payoffs $\{-1, 1, 3, 5\}$ and $P(o_k^{\text{act}} \mid \cdot)$ comes from Eq. (10). This preserves the Prisoner's Dilemma incentive structure at horizon 1 while allowing trust-building dynamics to emerge over longer horizons.

A.4 Transition dynamics

Type drift (B_k^{type}). Partner type is uncontrollable and drifts slowly:

$$P(s_k^{\text{type}\prime} = j \mid s_k^{\text{type}} = i) = \begin{cases} 1 - p_{\text{switch}} & j = i, \\ p_{\text{switch}}/(K_{\text{type}} - 1) & j \neq i, \end{cases} \tag{12}$$

with $K_{\text{type}} = 4$ and default $p_{\text{switch}} = 0.05$. Scheduled betrayal and mixed-volatility scenarios set $p_{\text{switch}} = 0$.

Stance dynamics (B_k^{stance}). Stance transitions are action-dependent on the focal agent's social action a:

$$P\big(s_k^{\text{stance}\prime} \mid s_k^{\text{stance}}, a\big)\,, \tag{13}$$

as specified below. Cooperation tends to build trust; defection damages trust; recovery from hostility is slower than collapse.

Table 4: Stance transition dynamics $P(s'^{\text{stance}} \mid s^{\text{stance}}, a)$ conditional on focal agent action.

Agent cooperates			
To \ From	Trusting	Neutral	Hostile
Trusting	0.90	0.30	0.05
Neutral	0.10	0.60	0.35
Hostile	0.00	0.10	0.60
Agent defects			
To \ From	Trusting	Neutral	Hostile
Trusting	0.10	0.05	0.02
Neutral	0.50	0.35	0.18
Hostile	0.40	0.60	0.80

These dynamics instantiate the asymmetry of trust: cooperation gradually builds trust (neutral $\rightarrow$ trusting in $\sim$ 3–4 steps), while defection rapidly damages trust (trusting $\rightarrow$ hostile in $\sim$ 2 steps); recovery from hostility is slow ($\sim$ 4–5 steps to neutral).

Own-action bookkeeping (B^{own}). The own-action factor updates deterministically from the executed social action:

$$P(s^{\text{own}\prime} = a' \mid s^{\text{own}}, a) = \mathbb{I}[a' = a]. \tag{14}$$

This factor is not a psychological hidden variable; it makes the agent's executed action available to the payoff likelihood in standard factorized form.

A.5 Preferences, priors, and policy priors

Observation preferences (C_k). There is no direct preference over partner actions:

$$C_k^{\text{act}} = [0,\, 0] \quad \text{over } \{\text{cooperate}, \text{defect}\}. \tag{15}$$

Payoff preferences ascend over $\{-1, 1, 3, 5\}$:

$$C_k^{\text{pay}} = \text{log\_softmax}\left(\frac{[-1, 1, 3, 5]}{\tau}\right), \tag{16}$$

where τ is a temperature parameter controlling preference sharpness. Instrumental motivation therefore enters through payoff observations and multi-step planning rather than direct action preferences.

Initial-state priors (D_k). Type, stance, and own-action priors are

$$D_k^{\text{type}} = \left[\tfrac{1}{4}, \tfrac{1}{4}, \tfrac{1}{4}, \tfrac{1}{4}\right], \tag{17}$$

$$D_k^{\text{stance}} = [0.2,\, 0.6,\, 0.2], \tag{18}$$

$$D^{\text{own}} = [0.5,\, 0.5] \quad \text{(binary task; uniform over graded levels in graded regimes).} \tag{19}$$

Policy priors (E_k). Policy priors are uniform over admissible action sequences at the configured planning horizon (no habitual bias toward cooperation or defection). For horizon 1 in the binary task, $E_k = [0.5, 0.5]$ over $\{\text{cooperate}, \text{defect}\}$; longer horizons enumerate all length-τ sequences with equal prior mass.

Partner selection across k is handled outside the per-partner POMDP by evaluating policy branches for each partner with precision-adjusted logits before executing the chosen action (Appendix B).

A.6 Partner implementation

Partner implementation follows Section 2.1. Four environment-side partners maintain latent type and stance and sample actions from the cooperation likelihood in Table 3, undergoing action-dependent stance transitions per Table 4. Partners do not perform variational inference or maintain affective precision estimates, isolating the focal agent's precision mechanism before extending to reciprocal active-inference partners.

B Affective Precision Update and Policy Selection

This appendix gives the β-update rule, policy-precision mapping, and partner-choice policy-selection procedure.

B.1 Categorical beta support

Each partner maintains a categorical posterior $q(\beta_k)$ over discrete levels $\{0.5, 0.67, 1.0, 1.5, 2.0\}$, updated each round from affective charge ϕ_k. The posterior is categorical rather than continuous; this keeps the inverse-β convention identifiable while preserving a compact implementation of the precision dynamics.

B.2 Persistence prior

The update proceeds in three steps. First, a persistence prior smooths the previous posterior via a tridiagonal transition matrix:

$$p(\beta_k) \leftarrow T \cdot q_{\text{prev}}(\beta_k), \tag{20}$$

where T is tridiagonal with persistence parameter 0.8 and reflecting boundaries, encoding the prior expectation that precision evolves slowly.

B.3 Charge-based pseudo-likelihood

Second, a pseudo-likelihood is constructed directly from affective charge, mapping positive charge to high precision (low β) and negative charge to low precision (high β):

$$\log \ell(\beta_\ell) = \phi_k \cdot \frac{1}{\beta_\ell}, \tag{21}$$

where β_ℓ ranges over the five support points. This is a hand-designed update rule (not derived from the POMDP generative model) chosen to satisfy monotonicity: strong positive charge reinforces low β, strong negative charge reinforces high β, and the effect scales with effective precision at each level.

B.4 Posterior update and policy precision

Third, the posterior is normalized:

$$q(\beta_k) \propto \exp(\log \ell(\beta_\ell)) \cdot p(\beta_k). \tag{22}$$

Policy precision is then set from the posterior mean as given in Eq. 3 of the main text. The β_k posterior is maintained as an external auxiliary tracker rather than a proper hidden state in the generative model, which simplifies implementation at the cost of losing the agent's ability to form prior expectations over its own future affective states (discussed in Section 4).

B.5 Centered cross-partner policy logits

In partner-choice regimes, precision must sharpen policies within a partner branch without creating artifacts when branches are compared across partners. The centered logit procedure is given in Eq. 4 of the main text. The agent samples or selects policies from the combined set of centered logits across all partner branches. Centering preserves the mean score of each partner branch while allowing γ_k to sharpen or flatten the within-partner policy distribution, preventing low-precision scaling from making uniformly poor partner branches spuriously attractive in cross-partner comparison.

B.6 Ablation variants

The four affect-runtime variants are defined in Section 2.5. For completeness: the no-affect control disables the β tracker entirely and fixes $\gamma_k = \gamma_{\text{base}}$; the tracked-only ablation updates $q(\beta_k)$ normally but fixes $\gamma_k = \gamma_{\text{base}}$, allowing the precision estimate to be analyzed without influencing action selection; the default partner-local variant applies the full update pipeline described in B.1–B.4 and sets γ_k from each partner's posterior mean; and the shared-β ablation maintains a single pooled β state updated from all partner evidence while keeping per-partner POMDP beliefs intact, with all partner branches inheriting precision from the shared expected β.

C Simulation Protocols and Metrics

This appendix summarizes simulation conditions, replication counts, schedules, and metrics.

C.1 Simulation conditions

All reported experiments use one focal active-inference agent with four partners ($K = 4$), per-partner generative models, and environment-side partner policies (Appendix A). Conditions differ by payoff regime, assignment mode, and scheduled partner changes.

Binary random-assignment baseline. In the binary trust game (Table 1 in the main text), partners are assigned randomly each round and the agent chooses cooperate or defect. Stochastic partner-type drift uses $p_{\text{switch}} = 0.05$ unless disabled for scheduled-switch scenarios.

Open graded decision regime. The graded regime replaces the binary action with a discretized investment level while keeping the same latent type–stance structure. This is the primary deployment-dissociation setting (Section 3.3): affect, no-affect, and tracked-only variants are compared on policy entropy and cumulative payoff.

Partner-choice regime. The agent first selects a partner, then selects an action toward that partner each round. Partner selection is where engagement reorganization and social-allocation readouts are defined (Section 3.4).

Focal-switch locality probe. A partner-choice graded environment with type volatility disabled ($p_{\text{switch}} = 0$) and a single scheduled stance switch on partner P0 tests whether local versus shared β tracking preserves the surprise-over-reward dissociation (Section 3.1).

Abrupt betrayal. A partner-choice graded environment with scheduled stance betrayal on a previously cooperative partner tests post-switch reallocation, entropy, payoff, and joint accuracy (Section 3.5).

Precision-dynamics parameter variation. Computational profile variants (low-gain, high-gain, and cautious-prior profiles) vary β_k amplitude and stability in the graded decision regime (Appendix D).

Extended profile experiments (Exp A–D). Parameter sweeps and factorial designs vary precision gain α and prior over model fitness to define computational trust-calibration profiles in open graded, betrayal, partner-choice, forgiveness, and mixed-volatility environments (Section 3.6).

C.2 Seed counts and episode lengths

Unless otherwise stated in the main text, episode length is 200 rounds. The abrupt-betrayal condition uses 120 rounds because post-switch readouts dominate interpretation. Profile-scale and mixed-volatility tables use 20 seeds per condition. Central behavioral comparisons use 30 seeds per condition. Exploratory diagnostic runs used three seeds per variant and were used only for implementation validation; all main-text claims are based on the confirmation-scale or profile-scale runs reported in the relevant sections.

Representative configured scales:

Table 5: Representative episode lengths and replication counts by experiment family.

Experiment family	Setting	Rounds	Seeds
H0/H2 open graded deployment	graded, random or open choice	200	30
H1 model fitness	graded / reliability probes	200	30
H3 focal-switch locality	partner choice, single switch	100	30
H4 partner choice	partner choice	200	30
H5 abrupt betrayal	partner choice, scheduled betrayal	120	30
H6 perturbation dynamics	graded perturbations	200	30
Exp A–D profiles	sweeps / factorial / forgiveness / mixed volatility	200	20

Supplementary materials. The supplementary repository contains the simulation environments, affect-runtime variants, experiment configuration files, analysis scripts, and figure-generation code used for the reported experiments.

C.3 Betrayal schedule

The canonical abrupt-betrayal protocol (H5 `betrayal_choice`) uses graded payoffs, agent-chosen partners, four partners with fixed initial types and stances, and no stochastic type switching ($p_{\text{switch}} = 0$). Partner P0 begins as a cooperator in a trusting stance. At round 31, P0 undergoes a scheduled stance switch to hostile while other partners retain their initial configurations. Primary readouts are post-switch cumulative payoff, policy entropy, partner reallocation, and joint partner-type accuracy over the remaining episode.

The profile-scale betrayal environment used in Exp A and Exp B applies the same single-switch structure within 200-round episodes for recovery-time and quadrant metrics. Exp C (forgiveness) extends this schedule: rounds 1–80 cooperative baseline, rounds 81–120 betrayal on P0, rounds 121–200 reversion to cooperative/trusting on P0.

C.4 Mixed-volatility schedule

The mixed-volatility environment (Exp D; Section 3.6) places four partners with heterogeneous volatility in the same partner-choice episode over 200 rounds:

- **P0:** stationary cooperator (no scheduled change).
- **P1:** stationary exploiter (no scheduled change).
- **P2:** abrupt shift—cooperative/trusting pre-switch, hostile/exploiter from the scheduled switch onward (round 101 in the implemented scenario).
- **P3:** gradual drift—stance and type transitions spread across mid-episode rounds (neutral stance from round 51; hostile/exploiter shift from round 151 in the implemented scenario).

Stochastic type switching is disabled ($p_{\text{switch}} = 0$). Conditions compare default, low-α, high-α, and no-affect variants at 20 seeds each.

C.5 Partner-choice allocation metrics

Partner-choice readouts quantify how precision reshapes *who* the agent engages, not only how it acts toward a fixed partner.

Selection share. Per-partner selection rate is the fraction of rounds on which each partner is chosen. Partner-choice reallocation is summarized qualitatively in Section 3.4.

Selection concentration. The Gini coefficient of the partner-selection distribution summarizes how concentrated engagement is across the social portfolio (used in profile Exp A and Exp B).

Post-betrayal allocation. Post-switch windows report reallocation away from the switched partner and toward alternatives, including post-betrayal P0 selection and high-investment rates in profile analyses.

Policy entropy under choice. Mean q_π entropy in partner-choice regimes captures how sharply the agent commits to partner–action policies (reported comparatively for affect versus no-affect controls).

C.6 Entropy, payoff, and accuracy readouts

Total payoff. Cumulative focal-agent payoff over the episode (or over defined post-switch windows for betrayal analyses).

Policy entropy. Mean q_π entropy ('q_pi_entropy' in logged outputs) measures action-commitment sharpness; lower entropy indicates sharper policy selection under precision modulation.

Joint partner-type accuracy. Mean joint correctness of inferred partner type ('mean_joint_accuracy') summarizes epistemic state inference; it is reported separately from payoff because precision modulation can change deployment and accuracy independently of cumulative reward.

β_k **range.** Per-partner or episode-aggregated range of β_k support summarizes precision-dynamics amplitude in profile and perturbation tables.

Model-fitness dissociation. Partner-local correlations between β_k-derived precision and action surprisal versus realized payoff test whether the confidence signal tracks predictive reliability rather than partner value (Section 3.1).

Mixed-volatility discrimination metrics. The discrimination index is the Pearson correlation between per-partner β_k trajectory and per-partner behavioral stability (higher indicates better separation of stable versus shifting partners). The P0 false-positive rate is the rolling rate at which engagement with the stationary cooperator P0 falls more than 15% below its early baseline despite P0 never changing.

Betrayal timing metrics. Recovery time and related latency readouts measure rounds after a switch until selection or payoff returns toward baseline thresholds (profile Exp A/B and betrayal analyses).

C.7 Replication summary

Central behavioral comparisons in Section 3 use 30 seeds per condition. Profile-scale and mixed-volatility analyses use 20 seeds per condition. The model-fitness dissociation (Section 3.1) uses active-encounter partial correlations to compare precision–surprise against precision–payoff associations, with shared β included as the relational-specificity ablation.

D Extended Profile and Robustness Results

This appendix reports extended parameter sweeps, profile analyses, and robustness diagnostics that support the main-text results.

D.1 Precision-dynamics parameter variation

We first vary the amplitude and stability of β_k dynamics to test how precision dynamics affect action commitment. These variants are computational controls rather than diagnostic categories.

Table 6: Precision-dynamics parameter variation in the graded decision regime.

Variant	β_k range	Total payoff	Policy entropy
affect	1.32	1851.3	8.6
low-gain	0.24	1840.5	8.9
high-gain	1.48	1859.0	8.1
cautious-prior	1.48	1877.7	8.4

Flattening the precision signal produces the clearest behavioral change. The low-gain variant compresses β_k range to 0.24 and raises entropy to 8.9, near the no-affect and tracked-only baselines. Expanded-dynamics variants produce wider β_k movement, but payoff does not follow range monotonically. The parameter sweeps below therefore vary gain and prior structure directly.

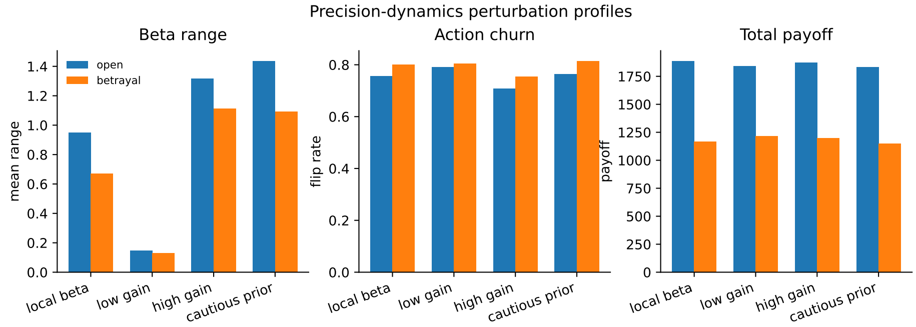

Fig. 5: Precision-dynamics parameter variation profiles in the graded decision regime. Variants compare precision-dynamics settings in the graded decision regime.

D.2 Alpha sweep

Sweeping α across $\{0.05, 0.1, 0.3, 0.5, 1.0, 2.0, 4.0, 8.0\}$ tests whether precision gain controls reactive social confidence across metrics. β_k movement range increases monotonically with α in the betrayal regime:

Table 7: β_k range by precision-gain α (betrayal regime, 20 seeds).

α	Mean β_k range
0.05	0.147
0.1	0.182
0.3	0.326
0.5	0.416
1.0	0.580
2.0	0.862
4.0	1.051
8.0	1.218

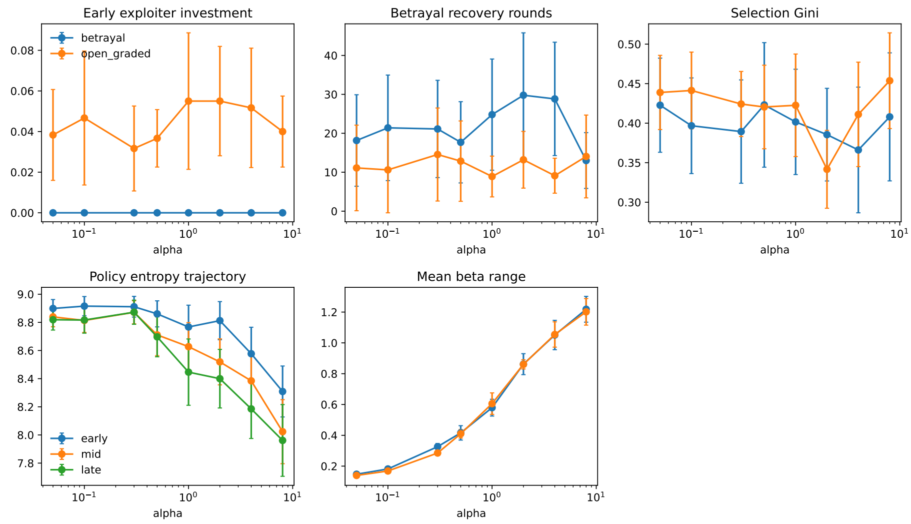

Fig. 6: Precision-gain sweep across open graded and betrayal environments (20 seeds). Higher α expands β_k range and reshapes engagement readouts without monotonic payoff ranking.

D.3 Profile quadrant profiles

Crossing prior belief (naive versus cautious) with α (low versus high) defines four behavioral profiles tested in the betrayal environment (20 seeds). Table 8 gives key metrics.

Table 8: Behavioral metrics by profile (betrayal environment, 20 seeds). Default agent shown for reference.

Profile	Payoff	Recovery	Gini	β range	Trust asymmetry
Anxious-reactive	1932	26.5	0.338	1.084	0.98
Hypervigilant	1951	20.1	0.423	0.892	1.65
Naive-stubborn	1894	21.3	0.304	0.364	1.19
Avoidant-rigid	1889	14.9	0.357	0.322	1.21
Default	1991	23.0	0.423	0.893	1.68

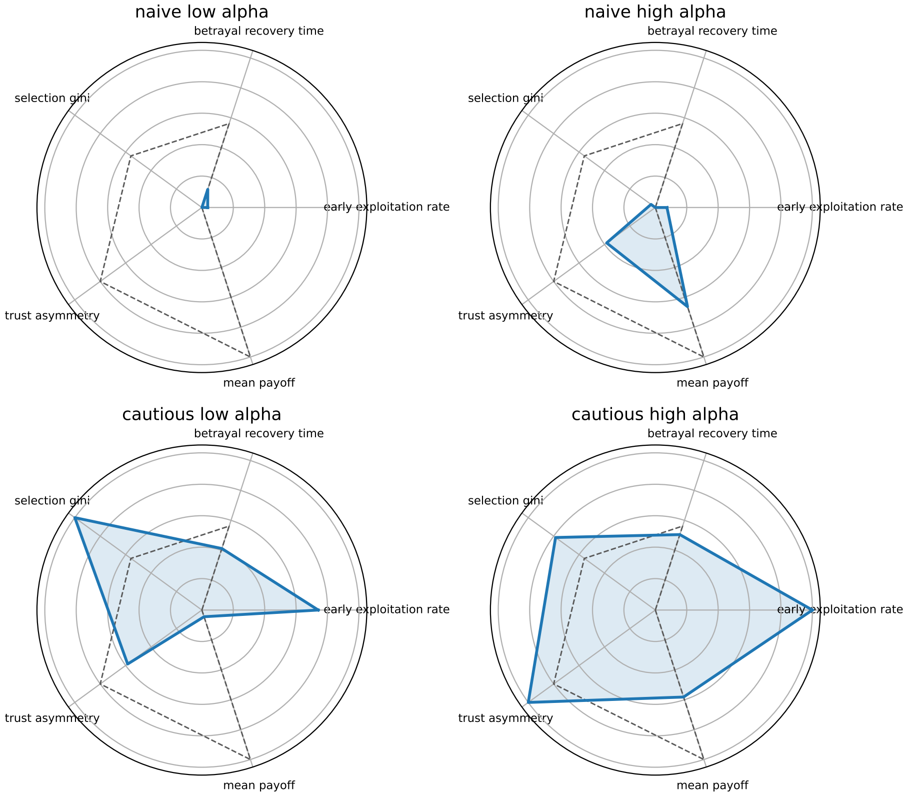

Fig. 7: Gain-prior profile quadrants from the prior-by-α factorial (betrayal environment, 20 seeds).

D.4 Forgiveness and trust repair

The forgiveness paradigm uses a 200-round partner-choice episode: cooperative baseline (rounds 1–80), betrayal on P0 (rounds 81–120), then reversion to cooperative/trusting behavior (rounds 121–200). Readouts include reengagement latency, post-return selection, payoff recovery, and β_k recovery trajectories for the reverted partner.

Table 9: Forgiveness and trust-repair metrics by precision profile (20 seeds). Reengagement is the post-repair P0 selection rate; payoff recovery compares late post-repair payoff with the late pre-betrayal baseline.

Variant	Reengagement	Payoff recovery	Latency	β_k range
cautious-high-α	0.518	0.996	5.5	0.971
cautious-low-α	0.630	1.014	1.5	0.287
default	0.475	1.019	3.0	1.012
naive-high-α	0.560	1.005	2.2	1.100
naive-low-α	0.527	1.004	4.6	0.325
no-affect	0.593	1.033	1.7	–

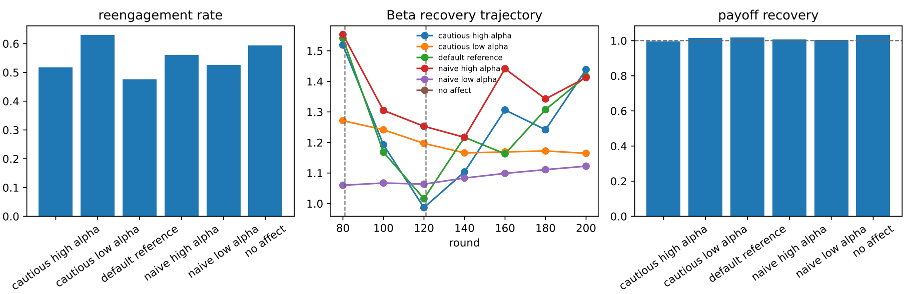

Fig. 8: Forgiveness/trust-repair diagnostic from the 20-seed profile analyses. Reengagement and payoff recovery do not collapse into a single monotonic gain effect: no-affect and cautious-low-α profiles reengage most often, while payoff recovery remains near baseline across profiles.

D.5 Mixed-volatility extended diagnostics

The primary mixed-volatility table and figure remain in Section 3.6. Extended Exp D diagnostics include per-partner β_k trajectories, rolling P0 selection concentration, and pairwise stability–precision correlations used to compute the discrimination index and P0 false-positive rate reported in the main text.